

Solutions to Problem Set 2

We make use of the following two basic facts, together with the comparison test to analyze improper integrals.

- If $a > 0$ then $\int_0^\infty e^{-ax} dx$ is convergent.
- $\int_1^\infty \frac{dx}{x^p}$ is convergent if and only if $p > 1$.

1. Observe that (by letting $x = t + 1$)

$$\int_1^2 f(x) dx = \int_0^1 f(t+1) dt = \frac{1}{2} \int_0^1 f(t) dt = \frac{1}{2}.$$

Similarly (by letting $x = z + 1$), we see that

$$\int_2^3 f(x) dx = \int_1^2 f(z+1) dz = \frac{1}{2} \int_1^2 f(z) dz = \frac{1}{4}.$$

We conclude that

$$\int_0^3 f(x) dx = \int_0^1 f(x) dx + \int_1^2 f(x) dx + \int_2^3 f(x) dx = 1 + \frac{1}{2} + \frac{1}{4} = \frac{7}{4}.$$

2. We write

$$f(x)^4 = f(x)^2 f'(x),$$

and use the substitution $u = f(x)$. It follows that

$$\int_0^1 f(x)^4 dx = \int_0^1 f(x)^2 f'(x) dx = \frac{1}{3} f(x)^3 \Big|_0^1 = \frac{1}{3} \left[1 - \left(\frac{1}{2} \right)^3 \right] = \frac{7}{24}.$$

3. (a) Observe that

$$e^{-x^4} \leq e^{-x} \quad \text{for all } x \geq 1.$$

Since

$$\int_0^\infty e^{-x} dx \quad \text{is convergent}$$

we conclude that

$$\int_0^\infty e^{-x^4} dx \quad \text{is convergent}$$

by the comparison test.

(b) Observe that

$$\frac{1 + e^{-x}}{x} \geq \frac{1}{x} \quad \text{for all } x \geq 1.$$

We conclude that

$$\int_1^\infty \frac{1 + e^{-x}}{x} dx \quad \text{is divergent}$$

by the comparison test.

(c) Observe that

$$\frac{x^2}{\sqrt{1+x^6}} = \frac{1}{x\sqrt{\frac{1}{x^6}+1}} \geq \frac{1}{\sqrt{2}x} \quad \text{for all } x \geq 1.$$

We conclude that

$$\int_1^\infty \frac{x^2}{\sqrt{1+x^6}} dx \quad \text{is divergent}$$

by the comparison test.

(d) Observe that

$$\frac{x}{\sqrt{1+x^6}} = \frac{1}{x^2\sqrt{\frac{1}{x^6}+1}} \leq \frac{1}{x^2} \quad \text{for all } x \geq 1.$$

Since

$$\int_1^\infty \frac{1}{x^2} dx \quad \text{is convergent,}$$

we conclude that

$$\int_0^\infty \frac{x}{\sqrt{1+x^6}} dx \quad \text{is convergent}$$

by the comparison test.

(e) Since the integrand is even, it suffices to look at

$$\int_0^\infty e^{-\sqrt{x}} dx.$$

(It is possible to actually compute the above integral explicitly, but I will give a solution based on the comparison test.) I claim that

$$e^{-\sqrt{x}} \leq \frac{256e^{-4}}{x^2} \quad \text{for all } x \geq 16.$$

To verify this inequality, it suffices to know that

$$t^4 e^{-t} \leq 256e^{-4} \quad \text{for all } t \geq 4.$$

The latter inequality follows from the fact that

$$\frac{d}{dt} t^4 e^{-t} = (4t^3 - t^4)e^{-t} \leq 0 \quad \text{for all } t \geq 4.$$

Since

$$\int_1^\infty \frac{1}{x^2} dx \text{ is convergent,}$$

we conclude that

$$\int_0^\infty e^{-\sqrt{x}} dx \text{ is convergent,}$$

by the comparison test, and consequently

$$\int_{-\infty}^\infty e^{-\sqrt{|x|}} dx \text{ is convergent.}$$

It is also fine to simply assert that we can choose $a, c > 0$ such that

$$e^{-\sqrt{x}} \leq \frac{c}{x^2}, \quad x \geq a,$$

by virtue of the Fact on page 29 from the notes for Lecture 3.

(f) Since

$$e^{-x^3} \geq 1 \text{ for all } x \leq 0,$$

we conclude that

$$\int_{-\infty}^\infty e^{-x^3} dx \text{ is divergent.}$$

4. Making the substitution $u = \ln x$ we see that

$$\int_e^\infty \frac{1}{x(\ln x)^p} dx = \int_1^\infty \frac{1}{u^p} du.$$

It follows that the integral in question converges if and only if $p > 1$ and the value of the integral for such p is $\frac{1}{p-1}$.

5. The integral clearly converges for $p \leq 0$, so we consider the case $p > 0$. Let us make the substitution $u = \ln x$ so that $x = e^u$ and $dx = e^u du$. Then we have

$$\int_e^\infty \frac{(\ln x)^p}{x^2} dx = \int_1^\infty u^p e^{-u} du.$$

Observe that

$$\frac{d}{du}(u^{p+2}e^{-u}) = (p+2-u)u^{p+1}e^{-u} \leq 0 \text{ for all } u \geq p+2.$$

Let us put

$$c = (p+2)^{p+2}e^{-(p+2)}.$$

It follows that

$$u^{p+2}e^{-u} \leq c \text{ for all } u \geq p+2,$$

and consequently

$$u^p e^{-u} \leq \frac{c}{u^2} \text{ for all } u \geq p+2.$$

We conclude that

$$\int_1^\infty u^p e^{-u} du \text{ converges}$$

by the comparison test. Therefore

$$\int_e^\infty \frac{(\ln x)^p}{x^2} dx$$

converges for all $p \in \mathbb{R}$.

It is also fine to simply assert that we can choose $a, c > 0$ such that

$$e^{-u} \leq \frac{c}{u^{p+2}}, \quad u \geq a,$$

by virtue of the Fact on page 29 of the notes from Lecture 3.

6. (a) Notice that

$$\frac{1}{\sqrt{t^3 + 1}} \geq \frac{t^{-\frac{3}{2}}}{\sqrt{2}} \text{ for all } t \geq 1.$$

It follows that

$$F(x) \geq \frac{1}{\sqrt{2}} \int_x^\infty t^{-\frac{3}{2}} dt = \sqrt{2} x^{-\frac{1}{2}} \text{ for all } x \geq 1.$$

Since $\int_1^\infty x^{-\frac{1}{2}} dx$ diverges, we conclude that $\int_0^\infty F(x) dx$ diverges.

(b) Notice that $F'(x) > 0$ for all $x \geq 0$ which implies that

$$F(x) \geq F(1) > 0 \text{ for all } x \geq 1.$$

It follows that $\int_0^\infty F(x) dx$ diverges.

(c) Observe that

$$e^{-t^2} \leq e^{-t} \text{ for all } t \geq 1.$$

It follows that for all $x \geq 1$ we have

$$F(x) \leq \int_x^\infty e^{-t} dt = e^{-x}.$$

Since

$$\int_0^\infty e^{-x} dx \text{ is convergent,}$$

we conclude that

$$\int_0^\infty F(x) dx \text{ is convergent,}$$

by the comparison test.